National University of Computer and Emerging Sciences

Lahore Campus

Data Communication & Networks(EE2007)

Sessional-II Exam

Date: November 2, 2026

Course Instructor(s)

Mr. Khalid Ijaz

Total Time (Hrs.):	**1**
Total Marks:	**15**
Total Questions:	**4**

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Roll No Section Student Signature

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Attempt all the questions.

CLO 3: Analyze the design principles involved in data link layer.

Q1: **Analyze** the throughput of a Pure ALOHA network in which 100 stations share a 1-Mbps channel, each frame is 1000 bits long, and each station transmits 10 frames per second. **[4 marks]**

CLO 3: Analyze the design principles involved in data link layer.

Q2: In the Figure shown below, the data rate is 10 Mbps, the distance between stations A and C is 2000 m, and the propagation speed is 2×10^8 m/s. Station A starts sending a long frame at time $t_1 = 0$ μs, while station C starts sending a long frame at time $t_2 = 3\mu s$. The frame size is long enough to guarantee collision detection by both stations. **[2+2 marks]**

Analyze:

a. The time when station C hears the collision (t_3).

b. The time when station A hears the collision (t_4).

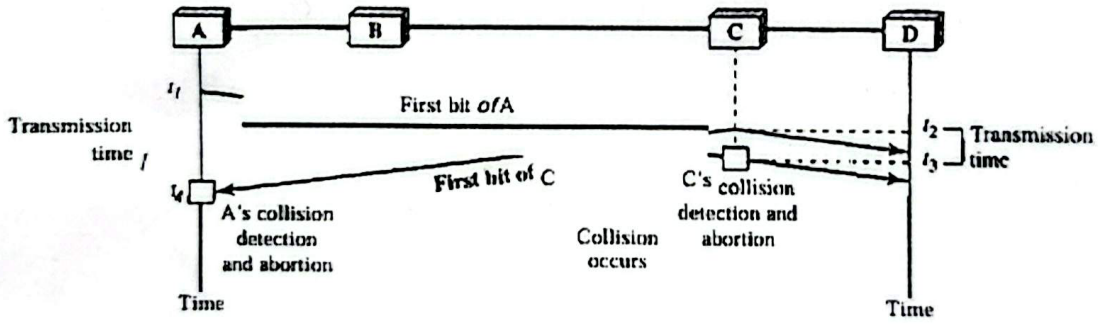


Figure 1: Collision of the first bit in CSMA/CD.

CLO 3: Analyze the design principles involved in data link layer.

Q3: A Wireless LAN employs the non-persistent CSMA protocol over a shared channel of 1 Mbps. Each data frame has a length of 1000 bits, and the propagation delay on the medium is 50 µs. **[2+2 marks]**

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1. **Analyze** the propagation time (a) as a fraction of the transmission time.
2. Using the approximate throughput efficiency expression for non-persistent CSMA:

$$S = \frac{G\,e^{-aG}}{1 + 2aG}$$

Where G is the offered load (frames per frame-time), **analyze** the throughput (S) when G=0.5.

CLO 4: Demonstrate the application of network layer protocols.

Q4: Consider the network shown in the figure, where Host H1, running Process P1, sends a packet to Host H2, running Process P2, through routers A, B, C, D, and E within the ISP's network. **Demonstrate** the routing table for Router E, indicating the appropriate next hop for forwarding packets toward Host H1 and Host H2. **[3 marks]**

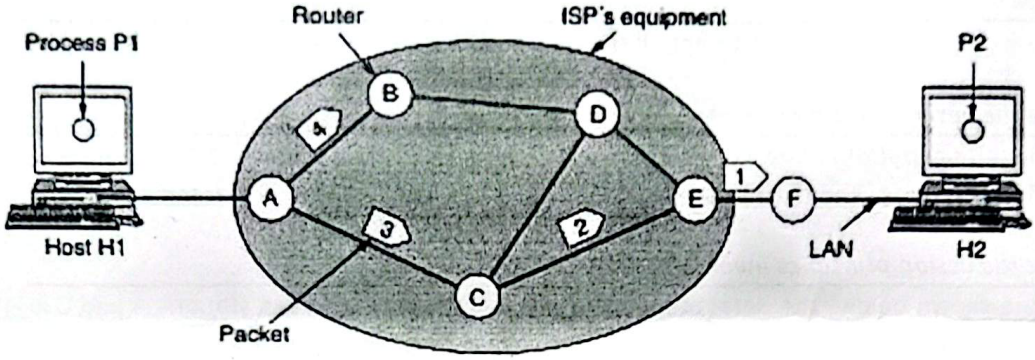


Figure 2: Routing within a datagram network.

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